

Nikola Milošević

PhD Researcher — Reinforcement Learning & Control Systems

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Research Interests

Foundations of reinforcement learning and control: theory of policy optimization, constrained and convex Markov decision processes, mathematical foundations of learning and control. Interest in applications in industrial process control and robotics. Broader interest in connecting these tools to models of learning in biological and artificial agents.

Education

2022 – present	PhD Researcher, Reinforcement Learning Neural Data Science Lab (advisor: Nico Scherf) at MPI for Human Cognitive and Brain Sciences. Affiliated with ScaDS.AI Leipzig. Developing tools from information geometry and constrained optimization to improve the reliability of deep reinforcement-learning algorithms.	MPI CBS Leipzig
2019 – 2021	M.Sc., Electrical Engineering & Information Technology Final grade 1.2 (German scale). - Thesis: <i>Segmentation of Ultra-High-Field MRI Brain Images for Multi-Parameter Mapping using Deep Learning</i> — a variational-autoencoder approach to unsupervised MRI segmentation. - Supervisor: Prof. Dr.-Ing. Gerold Bausch (HTWK), Kornelius Podranski (MPI).	HTWK Leipzig
2014 – 2018	B.Eng., Electrical Engineering & Information Technology Cooperative degree with Siemens (integrated vocational training).	HTWK Leipzig

Research Experience

2021 – 2022	Doctoral Research Stay Human and Machine Cognition Lab @ Uni Tübingen & MPI for Biological Cybernetics. - Studied reusable environment representations in human reinforcement learning. - Collaborators: Charley Wu, Philipp Schwartenbeck, Peter Dayan.	University of Tübingen
2019 – 2021	Research Assistant Built realistic simulations of high-resolution 3D MRI acquisitions to augment training data for deep-learning segmentation models.	MPI for Human Cognitive and Brain Sciences, Leipzig

Industry Experience

2022	AI Engineer Led a data-engineering project for the <i>Frankfurter Allgemeine Zeitung</i> and an ML project for the Saxony State Police (speaker diarization; a recommender for correcting automatic transcriptions).	Appsfactory, Leipzig
2018 – 2019	Embedded Software Engineer Controller software for smart solar-battery systems. Led two projects: a hardware-in-the-loop integration test suite and a communication interface to an external EV charger.	SENEC, Leipzig

2016 – 2018

Project Planning Engineer (Cooperative Degree)

Siemens, Leipzig

- Project planning for electrical substation control systems, alongside the integrated vocational training of the cooperative B.Eng. programme.

Teaching

present

Teaching Assistant, Advanced Deep Learning

Universität Leipzig

Gave lectures for master's students; deep reinforcement learning and deep latent variable model lectures.

2020 – 2021

Teaching Assistant, Systems Theory

HTWK Leipzig

Planned and led tutorials for third-semester students; final-exam preparation.

Awards

2021

Master's Thesis Award — *Förderpreis für angewandte Digitalisierung*.

Skills

Programming

Python (scientific computing, deep learning), C/C++ (mostly embedded).

Methods

Deep reinforcement learning, information geometry, constrained optimization, scientific computing, medical image analysis.

Languages

English (fluent), German (fluent), Serbian (fluent).